

SSDL-ILT: Efficient ILT utilizing a self-supervised deep learning model

Rui Xu*
Zhangjiang Laboratory
Shanghai 201204, China
xurui@zjlab.ac.cn

Junqi Yang
Zhangjiang Laboratory
Shanghai 201204, China

Haoxiang Jiang
Shanghai Jiao Tong University
Zhangjiang Laboratory
Shanghai 200240, China

Ming Fang
Zhangjiang Laboratory
Shanghai 201204, China

Abstract—Inverse lithography technology (ILT) is an advanced resolution enhancement technique that achieves mask optimization at the pixel level. However, application of ILT is hindered by time-intensive physical simulation. Herein, we propose an efficient ILT algorithm leveraging a deep learning model. A novel loss function is constructed to guide the model training in a self-supervised manner, eliminating the requirement of labelled data that might be nontrivial to acquire. The trained model outputs final mask patterns without further ILT optimization. Sub-resolution assist features (SRAFs) are generated automatically, the complexity of which can be adjusted during the training process to control mask manufacturability. The model was trained and validated on ICCAD-2013 CAD contest dataset. Better pattern fidelity and up to 12,000 times speedup are observed compared to other SOTA models. The trained model also shows good generalization ability to geometrically-different design patterns from another dataset, via a few-shot learning approach.

Keywords—ILT, self-supervised, deep learning, SRAF, few-shot learning

I. INTRODUCTION

Lithography is the optical process of creating pre-designed circuit patterns on a silicon wafer, and a mask is the template used to create these patterns. As the critical dimension of the design layouts gets increasingly smaller than the optical wavelength, strong imaging distortion/displacement effect occurs that has to be corrected in the process of mask design.

Optical proximity correction (OPC) is a major type of mask optimization technique. Mainstream OPC approaches include rule-based [1] and model-based OPC [2], [3]. Rule-based OPC corrects the mask pattern based on predefined rules or empirical formulas, which, however, are difficult to generalize to complex patterns. Model-based OPC simulates the imaging process of the optical system using physical models, allowing for more precise correction of distortions in mask patterns, but this method may suffer from limited solution space since the design patterns are optimized segment by segment [4]. Inverse lithography technology (ILT) [5] is an advanced mask optimization method used to address the challenges of shrinking feature size. ILT generates an optimized mask pattern by inversely solving the lithography process. Different from conventional OPC, ILT performs optimization at the pixel level, resulting in stronger flexibility and better fidelity. One of the major technical bottlenecks limiting the widespread application of ILT is the high computational demand and low efficiency.

With the development of artificial intelligence technology, deep-learning-based ILT algorithms reformulates the mask optimization problem as an image-to-image translation

problem. Namely, pre-trained neural network models predict mask patterns from target layouts directly, eliminating the need for solving complex physical models. Additionally, by utilizing GPU hardware acceleration, the computational efficiency is improved. Several state-of-the-art (SOTA) deep learning based ILT methods have shown improved performance compared to conventional ILT approaches, such as GAN-OPC [6], Neural-ILT [7], DevelSet [8], and A2-ILT [9]. A common workflow for these methods requires the pretraining of a deep learning model in a supervised manner, providing the model with thousands of design layout/mask pattern data pairs to learn the underlying mapping relationship. The output of the trained model serves as an initial guess for further ILT optimization to produce the final mask. This workflow is not computationally efficient. On the other hand, the supervised model training process may have limited fidelity or generalization ability when the training dataset is not adequate.

In this paper, we propose a self-supervised deep learning based ILT model (SSDL-ILT) for mask optimization with improved fidelity and efficiency. The contribution of our work is fivefold:

- (1) a novel self-supervised learning strategy is proposed utilizing a tailored loss function, which only requires design patterns as the training dataset;
- (2) end-to-end mask generation is realized without the need of further ILT correction;
- (3) sub-resolution assist features (SRAFs) are generated automatically, the complexity of which can be adjusted in the model training stage;
- (4) significant speedup is achieved at both the model training and inference stages compared to other SOTA deep-learning-based ILT models;
- (5) good generalization ability is demonstrated via a few-shot transfer learning approach.

The rest of the paper is organized as follows. Section II provides an introduction of the related lithography simulation background, mask optimization problem formulation, and the corresponding evaluation criteria. Section III describes in detail our proposed self-supervised deep learning based ILT algorithm. Section IV provides the experimental results, and conclusion is provided in Section V.

II. BACKGROUND

A. Forward lithography simulation

The forward lithography simulation computes the wafer image from the mask. It consists of an optical model and a photoresist model. The optical model produces an aerial image $\mathbf{I}$, which represents the light intensity distribution on the wafer surface. For a partially coherent optical system, the commonly used optical model is the Hopkins diffraction model [10]. By simplifying the model through singular value decomposition, a series of optical convolution kernels $\mathbf{h}$ can be obtained. The spatial image can be computed through the convolution operation between the mask image $\mathbf{M}$ and the optical convolution kernel $\mathbf{h}$:

$$\mathbf{I}(x, y) = \sum_{k=1}^N \omega_k |\mathbf{M}(x, y) \otimes \mathbf{h}_k(x, y)|^2, \quad (1)$$

where $\mathbf{h}_k$ is the k th optical kernel and ω_k is the corresponding weight; N is the total number of optical kernels used in the simulation, and here we set $N = 24$; $\otimes$ denotes the convolutional operation.

The photoresist model converts the continuous light intensity image $\mathbf{I}$ into a binary wafer image $\mathbf{Z}$. Here, we adopt a simplified threshold-based photoresist model. To smooth the wafer image allowing effective loss back-propagation in the deep learning model training process, a sigmoid function is further applied:

$$\mathbf{Z} = \frac{1}{1 + \exp[-(I - I_{th})]}, \quad (2)$$

where I_{th} is the light intensity threshold. Here, I_{th} is set to 0.225 consistent with other related works [6, 7].

B. Mask optimization problem formulation

Given a target layout $\mathbf{Z}_t$, the goal of mask optimization is to obtain an optimized mask $\mathbf{M}$ that could produce a wafer image as close to $\mathbf{Z}_t$ as possible, with acceptable runtime, low process variation band, and acceptable printability and manufacturability. The deep-learning-based mask optimization strategy is to replace the physics-based optimization process with a pre-trained surrogate model, which predicts the mask pattern directly from the target layout.

C. Evaluation metrics

Definition 1 (Squared L_2 error). Squared L_2 error is calculated by $\|\mathbf{Z}_t - \mathbf{Z}_{norm}\|_2^2$, given the target image $\mathbf{Z}_t$ and the wafer image $\mathbf{Z}_{norm}$. Here $\mathbf{Z}_{norm}$ denotes the wafer image under nominal dose and focus conditions.

Definition 2 (Process variation band, PVB). PVB denotes the summation of bitwise-XOR between two wafer images $\mathbf{Z}_{min}$ and $\mathbf{Z}_{max}$, which are generated under $\pm 2\%$ dose conditions, using the same settings as in the ICCAD-2013 CAD contest [11].

Definition 3 (Turnaround time, TAT): TAT is defined as the time elapsed during the model application for one mask optimization case, including the model prediction time and (if any) the ILT re-optimization time.

III. ALGORITHMS

A. Deep learning model for mask generation

We adopt a deep learning model to perform end-to-end mask optimization. An attention-based residual recurrent U-Net model (R2U-Net) is used which takes the target layout as input and outputs the corresponding mask pattern that would produce the target design. The attention R2U-Net model is a combination of R2U-Net [12] and attention U-Net [13]. The network architecture is shown in Fig. 1. In general, the U-Net model is composed of an encoder and decoder. The encoder down-samples the input target layout through convolutional and pooling operations to extract high-level features, whereas the decoder up-samples the extracted features and restore the image to the original resolution. Skip-connection [14] is applied at each up-sampling stage that fuses the features from the corresponding down-sampling stage to improve the model's memory of lower-level information. Attention mechanism [13] is added to the decoder so that the model learns the importance weights of each channel and attend to most related regions in the images to improve feature generation. Furthermore, the recurrent residual convolutional layers ensures better feature representation for training deep architecture. In the final layer of the decoder, a Sigmoid function is applied that restricts the output values to a range of 0-1, aligning with the value range of a binarized image.

B. Self-supervised learning strategy

Unlike traditional supervised deep learning models, the training process of our model is unsupervised, which is not guided by the data mismatch between predicted and referenced mask patterns. A novel loss function is constructed to incorporate physical guidance into the modeling training process. The loss function consists of three terms used to evaluate the quality of the generated masks. Given a series of layout patterns $(\mathbf{Z}_{t_1}, \mathbf{Z}_{t_2}, \dots, \mathbf{Z}_{t_{N_t}})$ similar in geometry and design to the target layout T , the neural network model ϕ_{NN} predicts the corresponding mask patterns

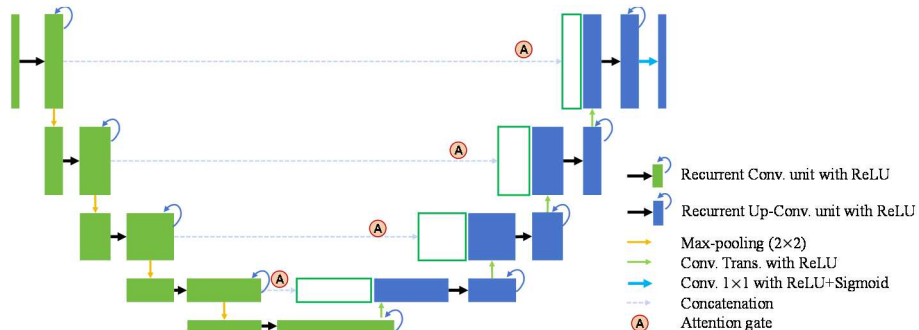


Fig. 1. Schematic of the attention R2U-Net model architecture.

$(M_1, M_2, \dots, M_{N_t})$, which are then subject to lithographical simulation ϕ_{sim} described in Section II to obtain the corresponding wafer images $(Z_1, Z_2, \dots, Z_{N_t})$. The first term $Loss_{L_2}$ in the loss function quantifies the similarity between the simulated wafer images and the corresponding design patterns:

$$Loss_{L_2} = \frac{1}{N_t} \sum_{i=1}^{N_t} (Z_i - Z_{t_i})^2, \quad (3)$$

where Z_t is the design pattern in the training set; N_t is the total number of design patterns used for model training; $\mathbf{w}$ is the parameter set of the neural network; Z is the simulated wafer image, and $Z = \phi_{sim}(\phi_{NN}(Z_t; \mathbf{w}); \text{norm})$. Here ‘norm’ indicates the dose and focus are at nominal conditions for the optical model.

The second term $Loss_{PVB}$ in the loss function evaluates the PVB of the generated mask:

$$Loss_{PVB} = \frac{1}{N_t} \sum_{i=1}^{N_t} [\phi_{sim}(\phi_{NN}(Z_t; \mathbf{w}); \text{min}) - \phi_{sim}(\phi_{NN}(Z_t; \mathbf{w}); \text{max})]^2, \quad (4)$$

where ‘min’ indicates the wafer image generated under the nominal focus and -2% dose condition, and ‘max’ indicates the wafer image generated under nominal focus and $+2\%$ dose condition.

Theoretically, the mask complexity and manufacturability can be quantified by the number of shots performed during the mask creation process. However, the shots calculation can be cumbersome and time-consuming. Herein we define a new evaluation metric $Loss_{com}$, which is the third term in the loss function to heuristically quantify the mask complexity:

$$Loss_{com} = \frac{1}{N_t} \sum_{i=1}^{N_t} [\text{sum}(Z_{norm_i}) / (H \times W)]^2, \quad (5)$$

where H and W indicate the image height and width in unit of total number of pixels. $Loss_{com}$ calculates the percentage of the image occupied by the mask pattern. Larger area of SRAF patterns will lead to higher value of $Loss_{com}$.

In the training process, the following total loss function is iteratively optimized:

$$\text{argmin}_{\mathbf{w}} \|\alpha Loss_{L_2} + \beta Loss_{PVB} + \gamma Loss_{com}\|_2^2, \quad (6)$$

where α, β, γ indicate the weight of the corresponding term. These weights can be tuned to generate mask patterns tailored to certain demand. Therefore, the model iteratively learns the physics behind mask optimization and generates masks that satisfy the defined evaluation criteria. The whole training process is in a self-supervised manner by providing the model with only certain amount of target layouts. The overall model training design is illustrated in Fig. 2.

During the training process, the backward calculation of the derivatives can be obtained as follows:

$$\begin{aligned} \frac{\partial Loss_{L_2}}{\partial \mathbf{w}} &= \frac{\partial Loss_{L_2}}{\partial \mathbf{M}} \frac{\partial \mathbf{M}}{\partial \mathbf{w}} \\ \frac{\partial Loss_{PVB}}{\partial \mathbf{w}} &= \frac{\partial Loss_{PVB}}{\partial \mathbf{M}} \frac{\partial \mathbf{M}}{\partial \mathbf{w}} \\ \frac{\partial Loss_{com}}{\partial \mathbf{w}} &= \frac{\partial Loss_{com}}{\partial \mathbf{M}} \frac{\partial \mathbf{M}}{\partial \mathbf{w}} \end{aligned} \quad (7)$$

$\frac{\partial Loss_{L_2}}{\partial \mathbf{M}}$, $\frac{\partial Loss_{PVB}}{\partial \mathbf{M}}$, and $\frac{\partial Loss_{com}}{\partial \mathbf{M}}$ can be calculated explicitly by deriving the lithographical model formula, the details of which can be referenced in [7]. $\partial \mathbf{M} / \partial \mathbf{w}$ can be obtained using the automatic differentiation feature provided by PyTorch [15]. The parameter set $\mathbf{w}$ of the neural network can then be updated by the following formula:

$$\mathbf{w}_{n+1} = \mathbf{w}_n + lr \frac{\partial Loss_{L_2}}{\partial \mathbf{w}}, \quad (8)$$

where lr represents the learning rate of the optimizer, and n represents the n^{th} iteration step.

Additionally, an early-stop mechanism is used during the training process to prevent overfitting and improve training efficiency. The training process stops when the loss function does not further reduce over the last ten epochs.

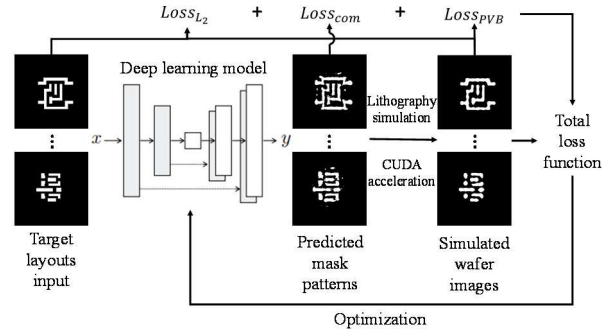


Fig. 2. Schematic of the self-supervised model training process.

C. Model inference

Once the model is adequately trained, it can be used for mask prediction given a target layout. If a target design is not in an image format (e.g. GDS files), it should first be converted into images, and the image size should be adjusted to match the input size of the model. The model's prediction result ϕ_{NN} needs to be converted into a binary image using a threshold value t .

$$\phi_{NN} = \begin{cases} 1, & \text{for } \phi_{NN} \geq t \\ 0, & \text{for } \phi_{NN} < t \end{cases} \quad (9)$$

Here we set $t = 0.5$. The entire self-supervised deep learning based ILT workflow is described in Algorithm 1.

Algorithm 1 The self-supervised deep learning based ILT workflow

Require: Target layout T for which a mask is to be generated, series of design patterns $\mathbf{Z}_t$ similar to T , lithography simulation model ϕ_{sim} , optical kernels $\mathbf{h}$, weights of the terms in the loss function α, β, γ , learning rate lr , maximum training epoch E_{max} .

Model training:

- 1: Initialize deep learning model architecture ϕ_{NN} and parameters $\mathbf{w}$;
- 2: Set $itr = 0$;
- 3: **repeat**
- 4: Mask prediction: $\mathbf{M}_{pred} \leftarrow \phi_{NN}(\mathbf{Z}_t; \mathbf{w})$;
- 5: Lithography simulation under different dose/focus conditions: $\mathbf{Z}_{norm}, \mathbf{Z}_{min}, \mathbf{Z}_{max} \leftarrow \phi_{sim}(\mathbf{M}_{pred}, \mathbf{h}; \text{norm}, \text{min}, \text{max})$;

6: Loss function calculation: $Loss_{L_2} \leftarrow \frac{1}{N_t} \sum_{i=1}^{N_t} [Z_{norm_i} - Z_{t_i}]^2$, $Loss_{PVB} \leftarrow \frac{1}{N_t} \sum_{i=1}^{N_t} [Z_{min_i} - Z_{max_i}]^2$, $Loss_{com} \leftarrow \frac{1}{N_t} \sum_{i=1}^{N_t} [\text{sum}(Z_{norm_i}) / (H \times W)]^2$, $Loss \leftarrow \alpha Loss_{L_2} + \beta Loss_{PVB} + \gamma Loss_{com}$

7: Gradient calculation: $\frac{\partial Loss}{\partial \mathbf{w}} \leftarrow (\frac{\partial Loss_{L_2}}{\partial \mathbf{M}} + \frac{\partial Loss_{PVB}}{\partial \mathbf{M}} + \frac{\partial Loss_{com}}{\partial \mathbf{M}}) \frac{\partial \mathbf{M}}{\partial \mathbf{w}}$

8: Network parameters update: $\mathbf{w} \leftarrow \mathbf{w} + lr \frac{\partial Loss}{\partial \mathbf{w}}$

9: Learning rate lr update based on cosine annealing;

10: $itr \leftarrow itr + 1$;

11: **until** $itr \geq E_{max}$ or $\min(Loss_{itr-10:itr}) > \min(Loss_{0:itr})$

Model inference:

12: $M \leftarrow \phi_{NN}(T; \mathbf{w})$

13: mask binarization: $M = \begin{cases} 1, & \text{for } M \geq 0.5 \\ 0, & \text{for } M < 0.5 \end{cases}$

model training and evaluation is a single NVIDIA RTX 4070 GPU card, and a 13th Gen Intel(R) Core(TM) i9-13900K@3.00 GHz CPU. The training dataset is obtained from the GAN-OPC paper [6], including 4000 synthesized target layouts based on the existing 32nm M1 layout topologies. The lithography model used for model evaluation is obtained from ICCAD 2013 contest [11].

A. Baseline performance test

For each design layout image in the training set, a smallest rectangle enclosing the patterns is cropped from the original image and mapped to the center of a 2048×2048 image. This operation is to center the patterns and improve optimization efficacy. The image size is then downsampled to 256×256 via an average pooling operation to reduce the computational cost and GPU memory requirement without losing necessary details. The output image is then interpolated to the original 2048×2048 size for calculation of Squared L_2 error and PVB.

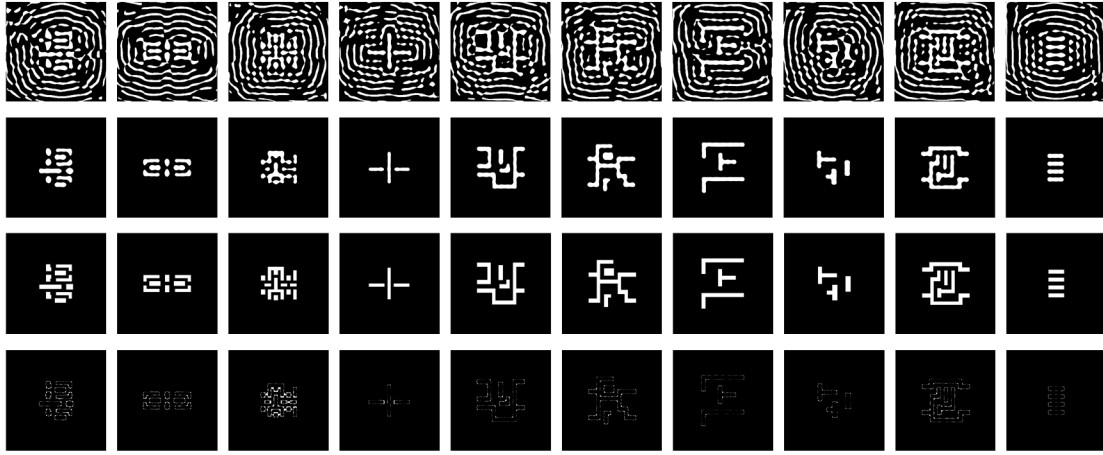


Fig. 3. The performance of our SSDL-ILT model on 10 test cases from ICCAD-2013 CAD contest. Rows from top to bottom indicate the predicted mask patterns, the corresponding simulated wafer images, the target layouts, and the absolute error between simulated wafer images and target layouts, respectively.

IV. EXPERIMENTAL RESULTS

The deep learning model is developed with PyTorch. The forward lithography model is adapted from the related module in OpenILT [16]. The entire workflow is deployed on CUDA to improve the solution efficiency. The hardware used for

α, β, γ in the loss function (6) is set to 1, 0.01, and 0, respectively. Adam is used as the optimizer and the learning rate is set to 0.0001. A cosine annealing learning rate decay scheduler is applied to improve loss convergence. The maximum training epoch is set to 50. The loss function converges after 32 epochs and the training takes about two

Table I. Performance comparison on 10 test cases from ICCAD-2013 CAD contest between our model and other SOTA models

Test case #	SSDL-ILT			Conventional ILT [17]			Re-optimization models								
							PGAN-OPC [6]			Neural-ILT 2.0 [7]			A2-ILT [9]		
	L2 (nm ²)	PVB (nm ²)	TAT (s)	L2 (nm ²)	PVB (nm ²)	TAT (s)	L2 (nm ²)	PVB (nm ²)	TAT (s)	L2 (nm ²)	PVB (nm ²)	TAT (s)	L2 (nm ²)	PVB (nm ²)	TAT (s)
1	34570	53793	0.031	49893	65534	1280	52570	56267	358	49817	55975	11	44477	49064	4.53
2	25991	42209	0.031	50369	48230	381	42253	50822	368	38174	52010	17	34031	40165	4.5
3	68138	84245	0.026	81007	108608	1123	83663	94498	368	89411	91357	10	88901	94850	4.54
4	7658	25760	0.032	20044	28285	1271	19965	28957	377	16744	29982	9	17464	24636	4.51
5	23597	57956	0.031	44656	58835	1120	44733	59328	369	45598	58900	11	35685	50664	4.53
6	27538	51164	0.031	57375	48739	391	46062	52845	364	43836	54969	10	41432	43516	4.52
7	11920	39441	0.031	37221	43490	406	26438	47981	377	20324	50542	16	21901	37688	4.51
8	8879	22001	0.032	19782	22846	388	17690	23564	383	13337	26353	15	15257	18548	4.48
9	32600	62556	0.031	55399	66331	1138	56125	65417	383	49401	68817	11	45885	56668	4.52
10	8259	16722	0.031	24381	18097	387	9990	19893	366	8511	20734	14	11121	14478	4.5
Average	24915	45585	0.031	44013	50900	789	39949	49957	371	37515	50964	12.4	35615	43028	4.51
Ratio	1.000	1.000	1.000	1.767	1.117	25634	1.603	1.096	12071	1.506	1.118	403	1.429	0.944	147

hours. 10 test cases provided by the ICCAD 2013 contest [11] are utilized to evaluate the model performance.

We show in Fig. 3 the predicted masks using our model, the simulated wafer images using the predicted masks, target layouts, and the absolute error between simulated wafer images and target layouts. The detailed performance results for each test case are shown in Table I, in comparison with conventional ILT [17] and other SOTA deep-learning-based ILT models. Table II shows the training time spent using our self-supervised training strategy and other supervised-learning-based models reported in the literature.

In general, deep learning based ILT models outperform conventional ILT by a large margin. Among deep learning ILT models, our model has the lowest L_2 error (on average 24915), achieving on average about 30% reduction compared to other SOTA models. The wafer images generated using our optimized masks show high fidelity to the target layouts, which indicates the effectiveness of our proposed self-supervised learning strategy. PVB reported by our model is also lower than all the other listed SOTA models (on average about 10% reduction) but A2-ILT (very close, though). Such PVB reduction results from the SRAF patterns generated by our model (Fig. 3). The most significant performance improvement using our model is the total TAT (on average 0.031s). Our model outputs the final optimized mask directly without further ILT optimization, achieving inference efficiency improvement of at least ~ 150 times compared to A2-ILT, and at most $\sim 12,000$ times compared to PGAN-OPC. Furthermore, as shown in Table II, the model training time is significantly reduced. Although the hardware used for training is not the same across different models, we expect a speedup of at least 5 times.

TABLE II. Model training time comparison

	SSDL-ILT	Re-optimization models		
		PGAN-OPC [6]	Neural-ILT 2.0 [7]	A2-ILT [9]
Training data size	4000	4000	4300	1000
Hardware	1 NVIDIA RTX 4070 GPU	1 NVIDIA Titan X GPU	1 NVIDIA Titan V GPU	1 NVIDIA RTX 3090 GPU
Training time (h)	2	10	19	25

B. Limited training data scenario

The baseline case in the previous section uses 4000 training samples to be roughly consistent with other SOTA models for comparison. Here we test the model's learning capacity in cases of limited training data, using 100, 500, and 2000 training samples, respectively, extracted randomly from the GAN-OPC training dataset. The results are shown in Table III. For comparison, we also train a supervised attention R2U-Net model, using the same network architecture and 4000 target layouts as training inputs, whereas the corresponding reference mask patterns are provided as well. As shown in Table III, our model's performance improves with the amount of the provided training data. The worst case scenario using only 100 training samples outperforms the supervised attention R2U-Net that uses 4000 training samples, and has

comparable results to other SOTA models, indicating the strong generalization ability of our self-supervised learning strategy. The model trained using 500 samples achieves better performance compared to other SOTA models. 2000 samples are enough to provide decent results comparable to 4000 samples as shown in Table I. In summary, our self-supervised learning strategy guarantees fidelity and robustness in the case of limited training data.

TABLE III. Effect of training data size on the model performance

Test case #	SSDL-ILT						Supervised attention R2U-Net	
	100 training samples		500 training samples		2000 training samples		4000 training samples	
	L_2 (nm ²)	PVB (nm ²)	L_2 (nm ²)	PVB (nm ²)	L_2 (nm ²)	PVB (nm ²)	L_2 (nm ²)	PVB (nm ²)
1	54415	53640	40226	54386	37648	54272	70836	56619
2	37683	43490	28218	42218	25940	41852	51819	48729
3	92939	72340	74839	82291	70842	81377	105215	64178
4	23192	27948	16912	29313	9503	27059	33527	23707
5	39426	56633	30741	58797	25570	58315	52156	61309
6	41365	45483	29600	50018	27245	51469	47784	54128
7	20241	38718	16382	40443	13010	39386	33947	46260
8	15700	20274	12542	21896	9822	21741	17812	24382
9	47908	56574	38284	60842	33184	61978	66007	62066
10	15652	16985	8935	16957	9013	16647	19441	19454
Average	38852	43209	29668	45716	26178	45410	49854	46083

C. Mask complexity adjustment

The weight γ in (6) controls the complexity of the mask, especially the generation of SRAFs. The baseline case is trained without any constraint of SRAF generation ($\gamma=0$). Here we show two more cases trained using $\gamma=0.005$ and 0.06 . The generated masks and corresponding wafer images are shown in Fig. 4. For $\gamma=0.005$, the third term in the loss function reduces the extent of SRAF generation, leaving only necessary patterns around the center design. The average L_2 error and PVB evaluated on the 10 cases are 25772 and 46167, respectively. For $\gamma=0.06$, SRAF generation is completely inhibited which improves the mask manufacturability. Similar L_2 error is reported (26636) but PVB is significantly larger (54477). Therefore, the major role of SRAF is the improvement of process window. In practical wafer production, the balance between PVB and mask manufacturability should be delicately determined.

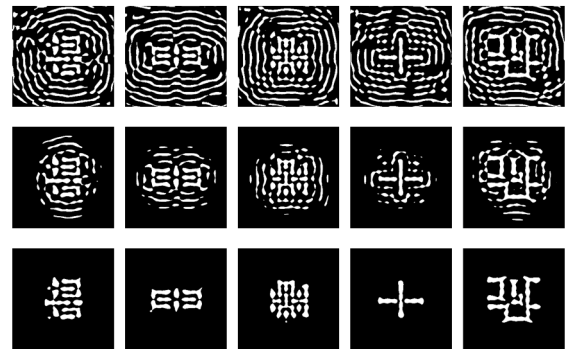


Fig. 4. Mask complexity adjustment demonstration. The rows from top to bottom are predicted masks using the trained model with $\gamma=0$, 0.005, and 0.06, respectively. For the sake of brevity, five test cases are shown.

D. Generalization ability test

We test the generalization ability of the model by applying the baseline model directly to 12 design patterns randomly extracted from the 2019 ISPD initial detailed routing contest, which are different in design and geometry from the training set. The results (5 out of 12 cases) are shown in Fig. 5. In general, good fidelity can be observed between simulated wafer images and design patterns (average $L_2=61460$). However, due to the existence of thicker and thinner pattern features than the training set, the wafer images obtained from the predicted masks show holes or missing/breaking thin lines in these regions. To further improve the model generalization ability, we adopt a few-shot transfer learning strategy using the pre-trained baseline model, by providing it with the remaining 7 design patterns from the 12 patterns as the training set. The network parameters are initialized to be consistent with the baseline model, and a smaller learning rate is applied. The model training is fast due to this small dataset, which completes in about two minutes after 300 epochs. The newly-trained model's performance is evaluated using the same five samples. As shown in Fig. 5, the holes in the previous wafer images are filled and feature resolution with small critical dimension is significantly improved compared to the original baseline model, which proves the strong adaptivity of our model (average $L_2=40687$). The few-shot transfer learning capability demonstrates potentially broad application scenarios of our model.

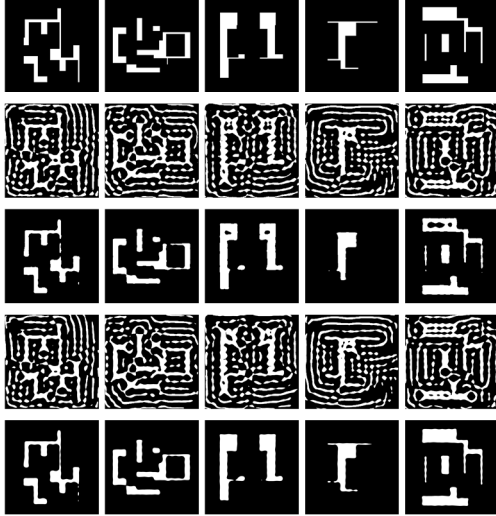


Fig. 5. Generalization ability test of the pre-trained model applied to design patterns extracted from ISPD 2019 initial detailed routing contest. The first row shows the five design patterns. The second and third rows are the predicted masks and simulated wafer images, respectively, obtained by the baseline model. The last two rows are results obtained by the few-shot transfer learning model.

V. CONCLUSION

In this paper, we propose a self-supervised deep learning model for direct mask prediction with high fidelity and efficiency. Critical lithography performance evaluation criteria is used to construct a novel physics-based loss function to guide the model training process. An attention-based R2U-Net architecture is applied to improve the model learning capability. Only hundreds of target layouts are required in the model training process instead of thousands of target/mask

data pairs as in supervised learning based SOTA models to achieve comparable results, which effectively alleviates the data acquisition burden. SRAF patterns of varying complexity can be generated automatically during the inference process, which effectively improves PVB. Experimental results on ICCAD-2013 CAD contest dataset prove better fidelity and efficiency of our model compared to other SOTA deep-learning-based ILT models. A few-shot transfer learning strategy applied on the ISPD 2019 initial detailed routing contest dataset shows that the pre-trained model can be efficiently adapted to predict masks from different layout structures with high fidelity.

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